MODULE ASSEMBLY AND PRESERVATION PROCEDURE

(Doc No. CT8- PRO-CM-014)

USA PROJECT UNIT 1 - TALLGRASS

Revision History				
Revisions	*Date*	*Descriptions*	*Prepared By*	*Approved By*
0	25-Mar-26	First Issue	Ms. Michelle Wilrutstien	Mr. Phisit Pew-La-Or

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1. SCOPE AND PURPOSE

This document covers the sequence of activities for erection, cleaning, preservation and installation procedure for Carbon/Alloy/Stainless Steels systems and to ensure the compliance to the above activities with the contractual applications in USA PROJECT UNIT 1 - TALLGRASS.

2. REFERENCES

LU-00130 : Quality-Requirements-for-Project
LU-00053 : Protection of Ends of Fabrication piping & skid assemblies
LU-00118 : Preservation, Labeling, Packing & Shipping Doc. Requirement
LU-00119 : Tagging & Packing Requirements for Interconnecting Pipe Hardware

3. MATERIALS AND EQUIPMENT'S

3.1 PRESERVATION

i) Desiccant bag

3.2 EQUIPMENTS

i) Borescope

4. PIPING SPOOLS ACCEPTANCE AND STORING

4.1 In general, all prefabricated spools delivered to BJCHI shall be stored indoor (closed area) with proper protection against moisture. All the prefabricated spools crates shall be identified with color and marking by inking, stamping, stenciling or pressing for clearly visible for identification purposes.

4.2 The storing of prefabricated spool piping shall be controlled by Warehouse.

4.3 Prefabricated spools open end shall be not opened until assembly works start.

5. MODULE CLEANING AND ASSEMBLY APPLICATION

5.1 All the assembly works shall be done at a closed area location.

5.2 Piping assembly area shall be separated from raw material cutting, piping fabrication, steel structural welding area.

5.3 BJCHI shall arrange a training program to educate assembly operator's, Welders regarding prevention of foreign objects debris. Eg: Check the button of worker's clothes before and after the work, check the items with the goods list before and after the work, etc. The training program shall mainly focus on the importance of prevention of foreign

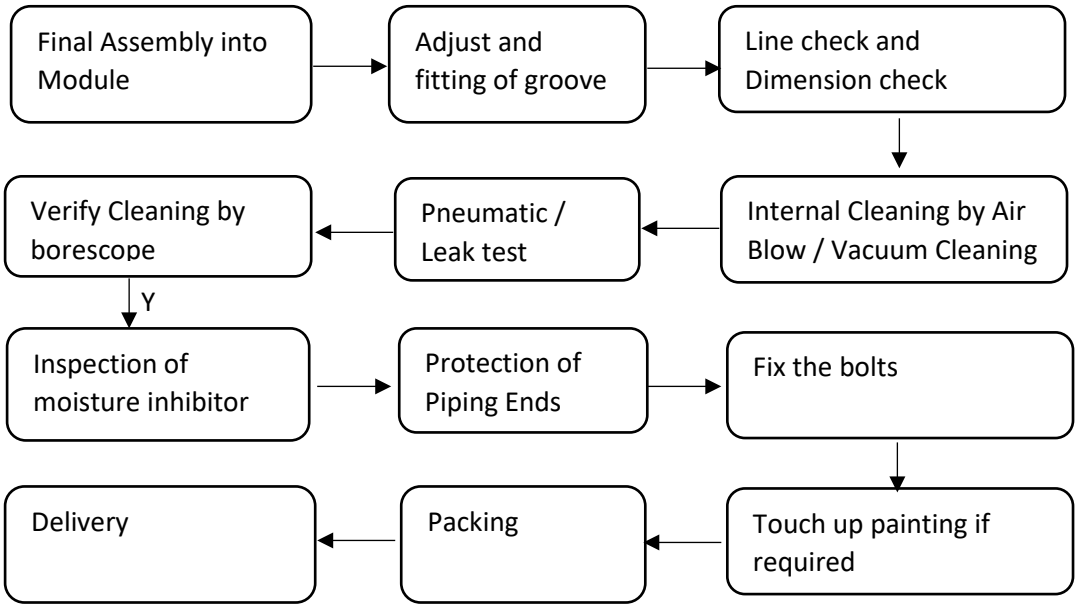
materials intrusion on the assembled equipment shall be known and fully understood by all the operators.

5.4 Before piping assembly, the re-inspection to the orientation of the equipment nozzle, embedment, penetrations, and steel structures relative to piping shall be conducted.

5.5 During assembly of spools, the method of adjustment shall be achieved through adjusting the packers under the pipe supports or piping hangers to the required elevation.

5.6 During assembly operation, the spool open ends will not be opened until the assembly starts. Spool open end will be opened only one side which will be assembled with connectin piping spools. Another open end of the spool still remains closed.

5.7 During the assembly welding process which includes fit-up, grinding, welding, fusing, etc., proper measures shall be taken to prevent the foreign object debris intrusion in advance. Dam plug shall be inserted inside the pipe one end away from the welding line. Dam plug shall be inserted in non-heat affect zone. Refer Attachment-3 for a sample picture of Dam plug. The dam plug application can apply only for straight pipe spools and will not complicate or risk to struggling during remove after completed application. Fabrication process such as fit-up, welding, etc., shall not be performed in the assembly area location.



5.8 During and after welding operation, 100% internal surface check shall be performed by QC inspector and supervised by CNBO inspection team by using Visual and borescope inspection after removal of dam plug from the pipe spool to ensure no foreign object debris exits inside the piping.

5.9 In case, if any foreign objects such as metal chip, dust, etc., stick on the spool internal surface, which is not easily accessible/reachable by operators such spools shall be dismantled from assembly and sent to the cleaning process to ensure internal cleanliness of the spool. Foreign object debris shall be removed by using pressurized air blowing or vacuum cleaning. 100% internal surface check shall be performed by Visual or borescope inspection to ensure no foreign object debris exits inside the piping. The foreign materials affected spools which can be easily accessible/reachable, such spools will not be dismantled from the assembly. Cleaning will be performed in the state of assembly position.

5.10 In case, if any spool has internal rust, which is not accessible/reachable by operators such spools shall be dismantled from assembly and sent to the cleaning process to ensure internal cleanliness of the spool such as removable of rust. For small rust portion, Cleaning shall be applied only the rust portion of the spool by wiping with clean white cloth by using WD-40. For heavy rust portion, the affected spools shall be cleaned by the acid cleaning process. After removable of rust, VpCI oil such as water or oil-based preservation shall be applied immediately to prevent rust occurring. VpCI shall be chosen as per piping system. Cleanliness criteria shall follow as per item 6.4 The rusted spools which can be accessible/reachable, such spools will not be dismantled from the assembly. Cleaning will be performed in the state of assembly position.

5.11 During welding, the welder shall ensure that valves and instruments are not included in the welding current circuit.

5.12 During the installation of spools and valves, more precaution shall be taken to prevent any damage from occurring during the handling, transportation, and erection.

5.13 After installation of spools and valves, adequate protection method shall be taken to prevent any damage from welding spatter, height falling, grinding, etc.

5.14 In case, if any spool needs to perform rework such as cut and re-weld due to dimension adjustment, such spool shall be separated from the assembly yard for rework operation. The connecting spools open end shall be covered immediately with flange cover to avoid debris, rust, etc.

5.15 After rework operation, spools shall be inspected 100% internal surface check by QC inspector and supervised by CIMTAS inspection team to ensure no foreign object debris inside the piping by visual or borescope inspection. After the internal check, suitable VpCI will be re-applied into the whole length of the pipe spool.

5.16 When the work shift ends, all the open ends shall be closed. Prior to close the open end, the operator shall verify the internal surface to ensure no foreign materials intrusion inside the pipe spools.

5.17 All piping shall be correctly supported during installation to prevent any permanent buckling, warps or bending.

5.18 After assembly completed, Pneumatic/leak test shall be performed to ensure there is no leakage in the system.

5.19 Before pneumatic/leak test completed, the whole module piping system shall be cleaned by pressurized air to ensure no foreign object debris exits inside the module piping system. In case, if any debris or rust exits, item 5.9 and 5.10 shall be followed.

5.20 Before Insulation operation, operators shall check and ensure that all the open ends are in a closed position to avoid foreign objects stick into the VpCI.

6. MODULE CLEANLINESS VERIFICATION

6.1 Each module piping system shall undergo 100% cleanliness verification inspection. Verification record shall be prepared and signed by the QC inspector. Verification record with photo/video proof shall be submitted to the client upon request. Refer sample cleanliness verification record in Attachment-2

6.2 Cleanliness Inspection shall be carried out with borescope.

6.3 In case, cleanliness inspection results failed due to a foreign object, rust, etc., item 5.9 and 5.10 shall be followed.

6.4 Cleanliness Criteria:

The following requirements should be checked before closing the pipe spools with flange cover caps.

i) There shall be no loose particles or foreign objects in the pipe spool.

7. PRESERVATION APPLICATION

7.1 After cleanliness verification, the Desiccant bag shall be placed on each end of blind flange cover plates. The desiccant bad shall be secured and tied at the blind flange cover plate to avoid movement during transportation.

7.2 Bag securing method shall allow its removal when the flange end cover plates are removed. Flange end cover plates shall be marked with the number of desiccant bags installed. Spool fabricator is responsible for ensuring the adequate design of desiccant bag application Refer to Attachment-1.

7.3 When desiccant is placed inside equipment (i.e.., spools, panels, junction boxes, etc.) an attention label shall be placed on the outside of the equipment stating “Silica bag inside”

7.4 Desiccant shall not be allowed to be in contact with the steel. Eg. Desiccant shall be fixed in the center of the cover flange with

7.5 The amount of desiccant (silica gel) to be applied has to be calculated for every pipe spool as per pipe dimension.

7.6 As quick as possible after QC Inspector checking the inner foreign particles, and application of moisture inhibitor, the pipe openings shall be covered and sealed to avoid rusting inside, entering foreign particles and damaging the weld edges or flange faces for the storage and transportation.

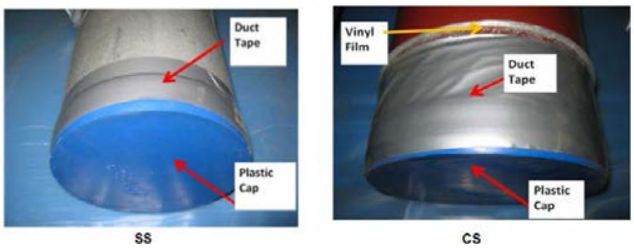
7.7 In case, if any spool preservation becomes dry due to environmental condition, VpCI shall be re-applied immediately. VpCI shall be checked inside the pipe spool periodically by operators.

7.8 BJCHI shall provide adequate protection of the product against deterioration and corrosion for a minimum of 6 months unless otherwise specified.

8. PROTECTION OF PIPE OPENINGS

8.1 Beveling edge

a) Cover with the Vinyl film for both CS/SS
b) Cover with the metal cap on the pipe end.
c) Put the duct tape over the plastic cap or steel cap and vinyl film to fix the cap and the pipe.



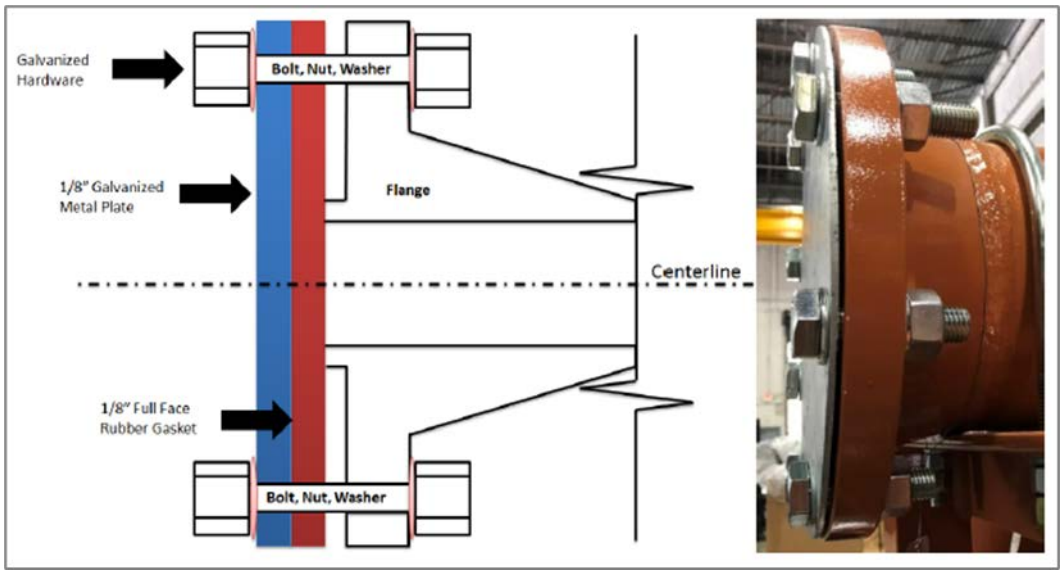
8.2 Flange ends

a) Coat the rust preventive over the flange surface with a brush (CS).

b) Rubber gasket to be installed

c) Cover with the metal cap on the flange

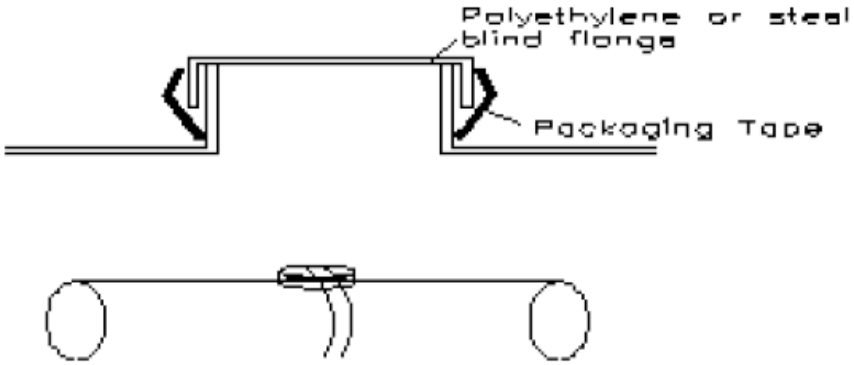
d) Bolt to be installed to fix the cap and the pipe



Flanged ends shall be closed by a full-face gasket and circular metal plate. The full-face rubber gasket shall be 1/8" thick, minimum, and shall have bolt holes matching the metal plate bolt holes for fastening. The rubber gasket shall be covered with a 1/8" thick galvanized circular metal plate. At least half of theflange's bolt holes shall be fastened and evenly spaced for an equal level of sealing compression. A minimum of 4 bolts, evenly spaced, shall be used for flanges containing 8 or fewer bolt holes. All hardware (bolts, nuts, washers, etc.) shall be galvanized. The minimum Bolt diameter shall be ½". Figure 2 gives an example of the flange end connection (metal plate painting not necessary).

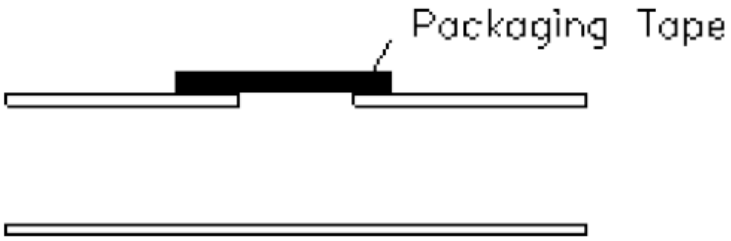
8.3 Boss and Nozzle stub

a) Cover with the Vinyl film on the opening (for CS)
b) Cover with polyethylene cap on the opening
c) Put the duct tape over the polyethylene cap and vinyl film to fix the cap and the pipe.
d) In case the boss and nozzle stub is too small to put the tape on, first seal them and tie a band around the mother pipe.



8.4 Hole

Tightly seal the hole with the plastic disc covered by vinyl film and duct tape. The hole up to 50mm diameter must be sealed two or three folds.



9. ATTACHMENTS

9.1 Attachment-1 Desiccant Bag Application Method

9.2 Attachment-2 Cleanliness Verification Record.

9.3 Attachment-3 Dam plug Sample Picture.

9.1 Attachment-1 Desiccant Bag Application Method



The amount of silica gel to be applied has to be calculated for every pipe spool, as follows:

Pipe dimension		Silica Gel in **gram** per 1 Meter	Pipe dimension		Silica Gel in **gram** per 1 Meter
2.5"	DN 65	2	18"	DN450	75
3"	DN 80	3	20"	DN500	93
4"	DN100	5	24"	DN600	134
5"	DN125	7	28"	DN700	187
6"	DN150	10	32"	DN800	247
8"	DN200	17	36"	DN900	314
10"	DN250	27	40"	DN1000	390
12"	DN300	38	44"	DN1100	473
14"	DN350	45	48"	DN1200	565
16"	DN400	59	56"	DN1400	773

The quantities above are calculated based on the assumption: per 1m3 Pipe volume = 500 gram silica gel are required.

9.2 Attachment-2 Cleanliness Verification Record

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REPORT No.:	
JOB NO.:	
PAGE NO.:	
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CLEANLINESS VERIFICATION RECORD

No.	DRAWING NO./REV.	SPOOL NO	DATE	VISUAL INSPECTION		REMARKS
				ACC.	REJ.	
1						
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EXAMINED BY: QC Inspector	APPROVED BY: QC Engineer
SIGN:	SIGN:
DATE:	DATE:

9.3 Attachment-3 Dam plug Sample Picture

